BRANDON SANTOR

emp1241.github.io

GEOSPATIAL RESEARCH | SCIENTIFIC SOFTWARE | U.S. MARINE CORPS VETERAN

Geospatial researcher and software developer building scientific tools from data preparation through the interface. Work spans tree-growth and climate analysis, hydrothermal-vent mapping, reconstructed Martian terrain, spatial evidence systems, and independent smartphone satellite-positioning research.

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Assistant and Software Developer | Torbenson Lab

March 2026-present | Paid appointment since June 2026

- Independently built the Tree-Ring Atlas Explorer (TRAE) in response to lab needs; its research use led to a paid appointment for continued software development and research support.
- Added reviewed imports, linked maps and graphs, climate matching, saved sessions, and exports that record sources and methods. Built tools for PRISM gridded climate data, NetCDF multidimensional files, ROXAS wood-anatomy outputs, and year-labeled WalDendro X-ray records.
- Used TRAE to derive summaries, organize study data, and prepare exports for a co-authored manuscript in preparation; exploratory screens support analysis but do not replace scientific validation.

Research Lead and GIS Developer | New World Screwworm (NWSW)

Fall 2026-present | Texas A&M GIS Practicum

- Lead the research synthesis: reconcile reporting sources, prepare reusable map boundaries, connect records to the areas they describe, and integrate the team evidence in a shared comparison table.
- Used Research Architect to generate the project coordination system, with versioned assignments and SHA-256 file fingerprints that help detect changed or mismatched handoff files.
- Independently built a separate spatial evidence application: FastAPI serves the browser interface, PostGIS links evidence records to geographic features, and TypeScript with Leaflet supports mapped review.

Independent Esri GIS Technical Consultant and Instructor

February-May 2026

- Audited seven graduate GIS modules and rebuilt six workflows into 189 pages of illustrated technical manuals.
- Corrected software and projection problems, misaligned map grids, database-structure and geographic-join errors, licensing issues, and desktop-to-web failures.
- Built quality-control checkpoints and taught reproducible workflows that enabled the client to execute corrected analyses independently.

High School Baseball Coach

2013-2020 | Paid program coach

- Led coaching for nearly 100 student-athletes, developing baseball skills, teamwork, and discipline.
- Coordinated conditioning, equipment, and maintenance for the program.
- Mentored student-athletes and emphasized education throughout their high school experience.

U.S. Marine Corps | Infantryman, Acting Platoon Sergeant, Logistics NCO

2011-2013

- Managed logistical support, inventory, and resource allocation for a company-sized element of approximately 350 Marines.
- Coordinated personnel and mission-critical field operations under strict safety, accountability, and risk-management requirements.
- Applied disciplined decision-making, communication, and logistical problem-solving in high-pressure field environments.

TEKsystems | Network and Computer Systems Contractor

2007-2011

- Executed enterprise hardware and software deployments involving 100 or more devices per project during nationwide infrastructure upgrades.
- Independently diagnosed complex computer and network issues across changing client environments.
- Installed, configured, and supported systems while coordinating with local and remote technical teams.

SELECTED SOFTWARE AND RESEARCH

Mars Explorer | Team Lead, Database and 3D Systems

Team lead

- Lead a five-person team. Built a spatial database with 41 connected tables for rover positions, science observations, and source files; developed data pipelines, a web API, and a Cesium terrain system that turns orbital elevation and imagery into bounded 3D terrain.

Thermal Vent Atlas Explorer | Independent Creator and Developer

Sylvan Lab | Delivered for review and internal use

- Built an online and offline research atlas mapping 742 hydrothermal-vent records against seafloor depth and tectonic boundaries, while keeping researcher-added microbial context separate from source records.

COLE - Cost Of Living Explorer | Project Lead

Application design and feature integration

- Lead design and feature integration for a county-level comparison tool combining modeled household budgets, long-term temperature estimates, and optional hazard context, with visible assumptions and comparison of up to three counties.

PhalAInx | Independent Software Studio

Creator and principal designer

- Created Research Architect, which configures an AI assistant kit for each member of a human research team and connects their work through shared methods, context, review, and handoffs.
- Built High Command v0.5.6, a React/TypeScript prototype for one person coordinating multiple AI models. It is in alpha testing; Tauri 2, Rust, and SQLite are the selected desktop stack, with native integration and model coordination in progress.

The Deception Gap | Smartphone Satellite-Positioning Research

Independent continued research

- Expanded a class project into independent research, more than doubling the original collection to 781 retained readings from three smartphones and five crowd categories; led collection, statistical analysis, interpretation, and updated poster development.

NASA SUITS Challenge Proposal | Team Lead and Primary System Designer

2021-2022

- Led conceptual HoloLens 2 mixed-reality interface design for extravehicular activity, including local positioning, navigation, terrain and hazard overlays, and geological annotation; co-authored technical documentation submitted to NASA for review.

EDUCATION

Texas A&M University | B.S. Geographic Information Science and Technology | Expected May 2027 | Institutional GPA: 4.0 / 4.0

ITT Technical Institute | A.S. Computer Networking | 2007

HONORS AND SCHOLARSHIPS

Second Place - TxGIS Day Undergraduate Student Research Presentation Competition | 2025 | Health Geography GIS and statistical analysis

Galloway '29 Aggie Veteran Freedom Scholarship II | Two-time recipient, 2025-2027

Dennis W. Hughes '79 and Melinda Hood Hughes '81 Endowed Scholarship in Geography | Two-time recipient, 2025-2027

Richard A. and Mollie B. Williford Dean's Scholarship | College of Arts and Sciences, 2026-2027

TECHNICAL TOOLKIT

Python; JavaScript and TypeScript; PostgreSQL and PostGIS; FastAPI and Flask; Cesium and Leaflet; NumPy, pandas, Rasterio; ArcGIS Pro and Field Maps; QGIS and GDAL; React and Vite; scientific data import and export; smartphone satellite-positioning analysis; JSON Schema; SHA-256 file checks.

RESEARCH COMMUNICATION

Co-authored manuscript in preparation with Torbenson Lab; scientific data preparation, workflow development, and research presentation.